

SIH26103 · MoSPI

AI-Powered Predictive Analytics & Early Warning System

Turning MoSPI's PAIMANA infrastructure-project data into forward-looking risk intelligence for cost overruns, delays, and early intervention.

Smart India Hackathon 2026 · Theme: Smart Automation · Org: MoSPI (DIID)

Team Concept Walkthrough — Draft for internal review

The Problem

PAIMANA tracks the data. It doesn't yet predict the risk.

1,981

ongoing projects tracked

17

central ministries

₹42.78L Cr

revised cost (vs ₹37.13L Cr original)

Even with all this data, projects still slip through:

- Reported progress often lags ground reality
- Delays are noticed only after milestones are missed
- Cost overruns are reported, not forecast
- Comparing 1,981 dissimilar projects is meaningless without benchmarks
- Root causes stay hidden inside dashboards that just say "delayed"
- Officers can't manually prioritise thousands of projects

Goal: move from descriptive reporting ("what happened") to predictive, prescriptive monitoring ("what's likely next, and which project needs attention now").

Our Solution

An AI/ML risk-intelligence layer on top of PAIMANA's historical project data — scoring every project's cost & delay risk, explaining why, and flagging early.



Predict

Cost & time overrun probability from historical patterns



Score & Warn

Project-level risk score + early-warning trend, before delay is official



Explain

Top contributing factors per project, in plain language



Benchmark

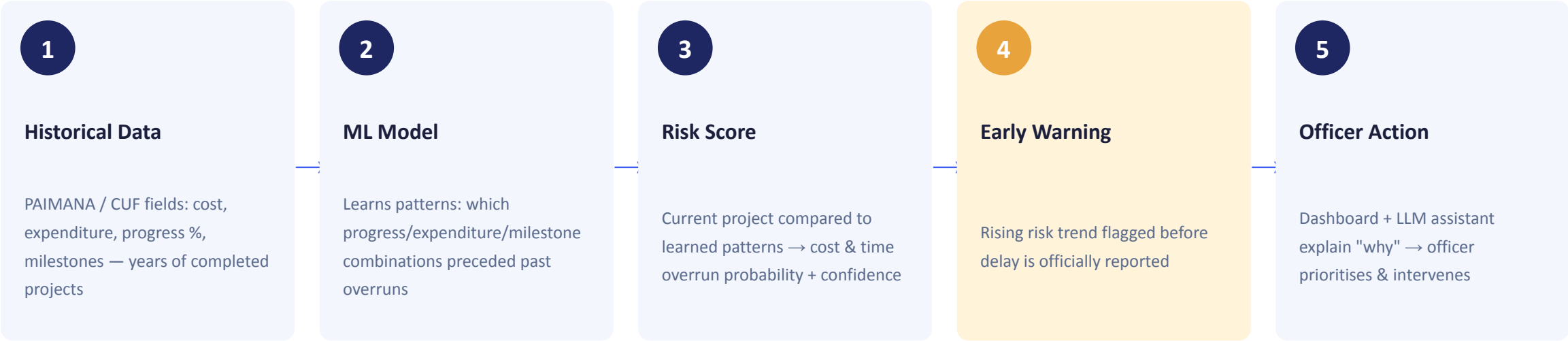
Compare each project against similar peers, not in isolation

Our differentiator

Real contractor-side ground research (from a road/bridge contracting background) into risk factors that never appear in MoSPI's official CUF form — payment-delay chains, subcontracting structure, utility/land issues. We test whether adding these to the model actually improves prediction over CUF fields alone — directly answering the PS's own question about CUF sufficiency.

How It Works

From historical patterns to an early-warning signal



Example risk trend (illustrative)



Risk crosses threshold in July — flagged 2+ months before typical milestone-miss reporting would have caught it.

What We're Building (Core Scope)

Six modules — matches MoSPI's official expected outcomes

1

Cost & Time Overrun Model

Predicts overrun probability from CUF-based historical fields

2

Risk Scoring + Early Warning

Live risk trend per project, alerts before delay is official

3

Benchmarking Module

"This project vs comparable peers" — not judged in isolation

4

Cost Escalation Driver Analysis

SHAP-based explainability: which factors are driving the score

5

LLM Project Intelligence Assistant

Officer asks "why is Project X flagged" in plain language

6

AI vs Statistics Benchmarking

Explicit comparison proving ML adds real value over conventional stats

CUF vs. Ground-Reality Factors

Our unique research input — what a contractor sees that the form doesn't capture

Already in CUF (official form)

- Approved & revised cost
- Expenditure to date
- Physical progress %
- Milestones & timelines
- Implementing agency

Our added ground-research factors

- Payment-delay → cash flow → manpower chain
- Subcontracting structure (main vs. petty contractor)
- Utility shifting / land clearance friction
- Design/scope change frequency
- Billing-vs-physical-progress mismatch signal

Experiment we run: train the model on CUF fields alone, then again with these added — and report the accuracy gap. That gap is our direct answer to the PS's own question about CUF sufficiency.

Scope: What We Build vs. What We Pitch as Vision

PHASE 1 — Build & Demo Now

- ✓ ML-based cost & time overrun prediction (CUF fields)
- ✓ Risk scoring + early-warning trend
- ✓ Benchmarking module
- ✓ SHAP-based driver analysis
- ✓ LLM assistant (open-source)
- ✓ AI-vs-stats comparison
- ✓ CUF-vs-added-factors comparison

PHASE 2 — Future Roadmap (Vision Only)

- ➔ Daily geo-tagged photo progress verification
- ➔ Billing-vs-progress anomaly detector (reframed, not an accusation engine)
- ➔ Officer visit/reminder escalation workflow
- ➔ Full role-based hierarchy workflow (18 roles)

Why split this way: PAIMANA's real data is monthly & tabular, not live photo feeds. Phase 2 has no dataset to demo in 36 hours — pitched as roadmap, not built.

Tech Stack

100% open-source, as required by the problem statement

Modeling	Python · scikit-learn · XGBoost / LightGBM	Cost & time overrun prediction, risk scoring
Explainability	SHAP	Per-project driver analysis, feature importance
LLM Assistant	Open-source LLM via Ollama / Hugging Face	Natural-language "why is this flagged" answers
Backend / API	FastAPI	Serves predictions & risk scores to dashboard
Database	PostgreSQL / DuckDB	Historical tabular project data
Dashboard	React + Recharts/Plotly (or Streamlit for demo speed)	Officer-facing risk dashboard & benchmarking view

From Reporting to Foresight

A predictive layer on PAIMANA that tells MoSPI not just what happened — but what's about to, and which project to look at first.

SIH26103 · Team Discussion Deck

Questions welcome — this is a working scope, not final.